

Flight 6 Substrate Specification v1.1

Status: Phase R1 deliverable for the substrate reconstruction cycle, post cross-AI review. This document specifies the computational substrate that implements the Step 2 Final commitments for Flight 6. Layer 2 (ChatGPT) and Layer 3 (Claude) cross-review converged on commit-readiness; this v1.1 revision incorporates the items both layers surfaced. Awaiting Mike's Step 4 commitment before implementation.

Scope: This specification covers the substrate code that produces .parquet telemetry. It does not cover Phase 4B extraction analysis, which is separately specified in the integrated Phase 4B specification.

Provenance: Refined from ChatGPT's Layer 2 draft specification with Mike's three arbitrations embedded as committed choices, additional architectural references, expanded implementation prohibitions reflecting today's substrate failure modes, and Section 16.2 items from Layer 2 and Layer 3 cross-review.

Changes from v1: Section 6 tick-pseudocode marks hash-emission points and explicit Q-application timing. Section 8 adds the realization-invariant verification granularity requirement. Section 10 cross-references the Q-timing anchor. Section 11 adds Term_Lambda as required column 25 (was implicitly computed in v1 but not logged) and adds required clipping-count reporting. Section 12 adds parity-scope clarification language. Section 14 specifies full per-(cell, tick) realization-invariant verification as default. Section 15 adds anti-inference prohibition. Section 16.2 records the eight items resolved through cross-review.

1. Governing principles

The substrate must implement the locked architectural commitments, not approximations of them. Every equation, every parameter, every computational element must trace to canonical record. Implementation that diverges from this specification — even in ways that appear simplifying or stabilizing — produces substrate substitution rather than substrate implementation.

The model produces agent-level telemetry sufficient for the analyses specified in the integrated Phase 4B specification. Telemetry-enabled execution must produce identical dynamics to telemetry-disabled execution; the parity check verifies this mechanically.

The cascade structure being implemented:

- Bases per cell: $b_i(t) = (v_i(t), u_i(t), r_i(t))$
- Activation drive per cell: $\Lambda_i(t) = F(b_i(t))$
- Probability chain: $\Lambda_i \rightarrow \text{drive}_i \rightarrow p_{\text{base}_i} \rightarrow p_{\text{act}_i}$
- Realization: $\text{PRNG_draw}_i < p_{\text{act}_i} \rightarrow \text{is_active}_i(t+1)$
- Population aggregates: $\rho(t), \Psi(t)$
- Spatial coherence per cell: $\Psi_{\text{local}_i}(t)$
- Slower-timescale base evolution: $Q: \Psi_{\text{local}_i} \rightarrow (\Delta v_i, \Delta u_i, \Delta r_i)$

Convention: $u = 1 - c$, where c is transition cost. Higher u indicates more favorable transition-cost conditions.

2. Architectural commitments and canonical sources

This specification implements commitments from three canonical sources:

Step A3 closure (drive function and parameters). The cascade activation drive uses:

$$\text{drive}_i = \alpha \cdot \Lambda_i + \beta \cdot \text{density}_i - \delta \cdot \text{density}_i^2 - \gamma$$

with parameters $\alpha = 4$, $\beta = 3$, $\delta = 4$, $\gamma = 4$. The nucleation floor $\eta = 0.01$ lifts p_{act} above p_{base} to prevent absolute zero activation probability.

Flight 5 closure (F-form implementations and weights). $F_{2_symmetric}$, the leading reformulation candidate for Open Element 14, takes the gated-weighted hybrid form:

$$\Lambda_i = (v_i \cdot u_i \cdot r_i) \cdot (0.33 \cdot v_i + 0.33 \cdot u_i + 0.34 \cdot r_i)$$

with symmetric weights $w_v = 0.33$, $w_u = 0.33$, $w_r = 0.34$. F_{LR} implements the Limiting Reagent form $\Lambda_i = \min(v_i, u_i, r_i)$. F_{baseline} implements the arithmetic mean as operational comparator.

Step 2 Final specification (25-column telemetry vector per v1.1, parity check protocol, three-probe structure, production parameters). The substrate produces per-agent per-tick rows containing the 25 specified telemetry columns, executes a parity check protocol verifying listen-only telemetry semantics, and supports the three-probe analytical structure (overcrowding penalty, F_{LR} ecological starvation with $F_{2_symmetric}$ cross-check, Q-A/Q-B fusion analysis).

Integrated Phase 4B specification (what telemetry must support). The substrate output enables the Tier 1 strict matching, Tier 2 coarser matching, Tier 3 interpretable regression with pre-specified interaction families, two-field empirical_result $\times$ structural_status classification, and cross-scale analysis specified for Phase 4B.

3. Global parameters

All parameters as Python constants in a single module:

Reproducibility

PRNG_SEED = 0x7A9B31C

Execution durations

TICKS_PARITY = 1000

TICKS_PRODUCTION = 3000

Grid sizes

GRID_PRIMARY = (20, 20)

GRID_VALIDATION = (40, 40)

Drive function coefficients (Step A3 closure)

ALPHA = 4.0 # Λ coefficient

BETA = 3.0 # positive density coefficient

DELTA = 4.0 # overcrowding coefficient

GAMMA_OFFSET = 4.0 # offset / threshold term

ETA = 0.01 # nucleation floor

Q operator

GAMMA_Q = 0.001 # continuous micro-integration rate

F_2_symmetric weights (Flight 5 closure)

W_V = 0.33

$W_U = 0.33$

$W_R = 0.34$

Base initialization range (committed arbitration)

$BASE_INIT_LOW = 0.6$

$BASE_INIT_HIGH = 0.9$

Telemetry epsilon thresholds for Phase 4B classification

$EPSILON_LAMBDA = 0.05$

$EPSILON_P = 0.05$

$EPSILON_SENSITIVITY = 0.01$

Naming convention: $GAMMA_OFFSET$ disambiguates the drive function's γ parameter from $GAMMA_Q$, the Q integration rate. $DELTA$ is the drive function's δ parameter (overcrowding coefficient); per-cell Q updates are named δ_v , δ_u , δ_r in lowercase. This prevents naming collision between the drive parameter δ and the Q update Δ .

4. Environment and dependencies

Required dependencies (pinned versions for reproducibility):

numpy==1.26.x

pandas==2.1.x or 2.2.x

pyarrow==14.x or later (for .parquet I/O)

mesa==3.0.x or later

hashlib (standard library)

Mesa version requirements. Mesa 3.x introduced API changes from earlier versions. `RandomActivation` is deprecated; use `model.agents.do("step")` and `model.agents.do("advance")` for synchronous two-phase update.

Alternative implementation note. If Mesa-specific abstractions create reproducibility complications, the substrate may be implemented as a pure NumPy grid engine that reproduces the agent dynamics exactly. If this alternative is taken, the substrate file must declare:

"NumPy substrate implementation of locked Mesa-equivalent dynamics. Verified to produce identical telemetry to Mesa implementation under matched PRNG seed."

with a verification artifact comparing outputs across both implementations under matched seed. No silent substitution between implementations.

Path handling. The substrate code must use relative paths, configuration-driven I/O, or environment-agnostic file handling. Hardcoded absolute paths or environment-coupled assumptions that break under directory relocation are explicitly prohibited. Path handling should be tested against directory moves before any production execution.

5. Initialization

5.1 Grid

Toroidal grid of dimensions specified by GRID_PRIMARY or GRID_VALIDATION. Moore neighborhood at each cell includes the 8 surrounding cells with torus wrapping. The torus wrapping is essential for the homogeneity assumption — agents at the grid boundary have neighbors via wrap-around, not zero-padding.

5.2 Base initialization (committed arbitration)

Each cell's bases are stochastically initialized independently:

$v_i(0) \sim \text{Uniform}(0.6, 0.9)$

$u_i(0) \sim \text{Uniform}(0.6, 0.9)$

$r_i(0) \sim \text{Uniform}(0.6, 0.9)$

Rationale for U(0.6, 0.9): This range produces evaluable Λ values for F_2_symmetric across typical cellular configurations. Specifically, with bases in [0.6, 0.9], the multiplicative term $v \cdot u \cdot r$ ranges approximately [0.216, 0.729], and the additive term ranges [0.6, 0.9]. F_2_symmetric's $\Lambda = \text{mult} \cdot \text{add}$ then ranges approximately [0.13, 0.66] — substantially above the $\epsilon_\Lambda = 0.05$ suppression threshold across the cellular population.

This preserves Flight 5 closure's commitment to $F_{2_symmetric}$ as repair candidate under conditions where F_2 produces meaningful activation drive. The prior physical runner produced $F_{2_symmetric} \wedge \in (0.016, 0.037)$, entirely below $\epsilon_\wedge$ — this represented either a specification error in initialization or an unexamined regime; either way, it contradicted Flight 5's assumption that $F_{2_symmetric}$ is evaluable.

Heterogeneity preserved. Stochastic initialization produces per-cell base variation, generating the spatial structure the cascade architecture depends on. Uniform initialization (every cell starting at identical values) is explicitly prohibited; it collapses spatial dynamics.

5.3 Activation initialization

After bases are initialized, initial activation states are drawn randomly:

$is_active_i(0) \sim \text{Bernoulli}(0.5)$

50/50 random initialization is the baseline. Other initial configurations (gradient, clustered, etc.) are not part of Flight 6 reconstruction; they may be relevant for future flights.

5.4 Randomness and reproducibility

Use one explicit PRNG stream per run:

```
rng = numpy.random.default_rng(PRNG_SEED)
```

The substrate must not mix Python's `random` module, NumPy's legacy `np.random`, and Mesa's RNG. If Mesa's RNG is used (which it may be for agent ordering), seed it explicitly with the same `PRNG_SEED` and log the seed value at execution start.

Reproducibility requirements:

- Telemetry-disabled parity run and telemetry-enabled parity run must produce identical PRNG state evolution
 - Production runs under same F form and same scale must reproduce under matched seed
 - Different F forms run under matched seeds share the same initialization PRNG state; divergence between F forms results from dynamics, not seed mismatch
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6. Per-tick semantics (synchronous two-phase update)

Each tick proceeds in the following order. The synchronous structure is essential: cellular activation transitions at tick t are computed based on neighbor states at tick $t-1$, then all cells advance simultaneously.

Timing anchors (v1.1): The pseudocode below explicitly marks (a) the state at which pre-Q base values are read and logged for telemetry, (b) Q-application timing relative to Ψ_{local} computation, and (c) hash-emission points for $\rho(t)$ and $\Psi(t)$ parity series. These anchors resolve telemetry timing ambiguity and establish the verification structure for Phase 4B residual analysis.

Tick t execution sequence:

1. **Read current bases [PRE-Q STATE — LOGGED TO TELEMETRY].** For each cell i , $v_i(t)$, $u_i(t)$, $r_i(t)$ are read into the per-cell computation. These are the pre-Q base values; columns b_{i_v} , b_{i_u} , b_{i_r} in the telemetry row for tick t record these values, not the post-Q values produced in steps 11-12.
2. **Compute per-cell Λ .** For each cell i , $\Lambda_i(t) = F(v_i(t), u_i(t), r_i(t))$ using the selected F form (Section 7). Computed from pre-Q bases.
3. **Compute local density.** For each cell i , $\text{density}_i(t) = (1/8) \cdot \sum \text{is_active}_j(t)$ over Moore neighbors j . Density is normalized to $[0, 1]$, the fraction of active Moore neighbors (committed arbitration).
4. **Compute drive components.** For each cell i :
 - $\text{Term_Lambda}_i = \alpha \cdot \Lambda_i(t)$
 - $\text{Term_Density_Pos}_i = \beta \cdot \text{density}_i(t)$
 - $\text{Term_Overcrowding}_i = -\delta \cdot \text{density}_i(t)^2$
 - $\text{Term_Offset} = -\gamma$ (constant across cells)
 - $\text{Drive_Raw}_i = \text{Term_Lambda}_i + \text{Term_Density_Pos}_i + \text{Term_Overcrowding}_i + \text{Term_Offset}$
5. **Compute probability chain.** For each cell i :
 - $p_{\text{base}_i} = \text{sigmoid}(\text{Drive_Raw}_i) = 1 / (1 + \exp(-\text{Drive_Raw}_i))$
 - $p_{\text{act}_i} = p_{\text{base}_i} + \eta \cdot (1 - p_{\text{base}_i})$
 - p_{act_i} clipped to $[0, 1]$ (numerical safety)
6. **Draw and determine next state.** For each cell i :

- $\text{PRNG_draw}_i \sim \text{Uniform}(0, 1)$
 - $\text{next_state}_i = 1$ if $\text{PRNG_draw}_i < p_{\text{act}_i}$ else 0
7. **Synchronous advance.** All cells update simultaneously: $\text{is_active}_i(t+1) = \text{next_state}_i$ for all i . This is the synchronous two-phase update; no cell sees a partially-updated grid.
 8. **Compute activation change.** For each cell i : $\text{ds}_i(t) = \text{is_active}_i(t+1) - \text{is_active}_i(t)$. The activation change is bipolar: -1 (deactivation), 0 (no change), +1 (activation).
 9. **Compute Ψ_{local} .** For each cell i : $\Psi_{\text{local}_i}(t) = \sum \text{ds}_i(t) \cdot \text{ds}_j(t)$ over Moore neighbors j . This is the activation-change correlation between cell i and its neighbors. When $\text{ds}_i = 0$, the contribution from cell i is zero.
 10. **Compute global aggregates [HASH-EMISSION POINT].**
 - $\rho(t+1) = (1/N) \cdot \sum \text{is_active}_i(t+1)$ — post-activation population rate for tick $t+1$
 - $\Psi(t) = (1/N) \cdot \sum \Psi_{\text{local}_i}(t)$ — transition-coherence at tick t , computed from $\text{ds}(t)$
 - These values are emitted to the parity-hash time series at this point. The hashed ρ series is post-activation; the hashed Ψ series is from transition correlation at the same tick. Column-name convention in optional global logs:
 $\text{rho_after_activation}[t+1]$ and $\text{psi_from_transition}[t]$.
 11. **Apply Q operator [Q-APPLICATION ANCHOR].** Q is applied after Ψ_{local} has been computed for tick t (step 9) and before tick $t+1$'s Λ computation (next iteration's step 2). For each cell i (committed arbitration: local Ψ_{local} input):
 - $\Delta v_i(t) = \gamma_Q \cdot \Psi_{\text{local}_i}(t)$
 - $\Delta u_i(t) = \gamma_Q \cdot \Psi_{\text{local}_i}(t)$
 - $\Delta r_i(t) = \gamma_Q \cdot \Psi_{\text{local}_i}(t)$
 12. **Update bases [POST-Q STATE].** For each cell i :
 - $v_i(t+1) = v_i(t) + \Delta v_i(t)$
 - $u_i(t+1) = u_i(t) + \Delta u_i(t)$
 - $r_i(t+1) = r_i(t) + \Delta r_i(t)$
 - Bases clipped to $[0, 1]$ (numerical safety). Per-tick clipping counts for v , u , and r are tracked and reported per Section 11. The post-Q bases become the pre-Q bases at the start of tick $t+1$'s step 1.
 13. **Record telemetry.** Write the per-cell per-tick telemetry row for tick t (Section 11), containing: the pre-Q base values from step 1, the Λ and drive components from steps 2-4, the probability chain from step 5, the realized state from step 7, the spatial

coherence from step 9, and the Q updates from step 11. The row records pre-Q bases and the post-Q $\Delta v/\Delta u/\Delta r$ applied to them; the post-Q bases appear as pre-Q bases in tick $t+1$'s row.

The tick ordering ensures telemetry rows contain a single coherent time-index: pre-Q bases that produced this tick's Λ , drive, probability chain, and activation transition, plus the Δ values applied in response to the resulting Ψ_{local} .

7. F-form implementations

All F-forms operate per cell on the current cell's bases. Λ_i is a per-cell quantity; there is no global Λ parameter.

7.1 F_LR (Limiting Reagent)

$$\text{Lambda}_i = \min(v_i, u_i, r_i)$$

Vectorized implementation:

$$\text{lambda_total} = \text{np.minimum}(\text{np.minimum}(v_grid, u_grid), r_grid)$$

7.2 F_2_symmetric (gated-weighted hybrid; Flight 5 leading candidate)

$$\text{Lambda_multiplicative} = v_i * u_i * r_i$$

$$\text{Lambda_additive} = 0.33 * v_i + 0.33 * u_i + 0.34 * r_i$$

$$\text{Lambda}_i = \text{Lambda_multiplicative} * \text{Lambda_additive}$$

Vectorized implementation:

$$\text{lambda_mult} = v_grid * u_grid * r_grid$$

$$\text{lambda_add} = W_V * v_grid + W_U * u_grid + W_R * r_grid$$

$$\text{lambda_total} = \text{lambda_mult} * \text{lambda_add}$$

No rescaling. No clipping unless numerical operations produce values outside $[0, 1]$ (which is theoretically possible if $\Lambda_{\text{multiplicative}} * \Lambda_{\text{additive}} > 1$ given high base values). If clipping occurs, log the count of clipped cells per tick.

7.3 F_baseline (arithmetic mean; operational comparator)

$$\text{Lambda}_i = (v_i + u_i + r_i) / 3$$

F_baseline is an operational comparator for parity checks and reference behavior. It is NOT an architectural candidate for Open Element 14. Phase 4 findings do not engage F_baseline as a substantive F-form claim.

7.4 F-form selection mechanism

The substrate accepts F-form as a model parameter:

$$f_form \text{ in } \{ "F_LR", "F_2_symmetric", "F_baseline" \}$$

Each production run logs its F_variant explicitly. The selection is per-run; a single execution does not switch F-forms mid-simulation.

8. Drive and probability chain

The drive function from Step A3 closure:

$$\begin{aligned} \text{drive}_i &= \alpha \cdot \Lambda_i + \beta \cdot \text{density}_i - \delta \cdot \text{density}_i^2 - \gamma \\ &= \text{Term_Lambda}_i + \text{Term_Density_Pos}_i + \text{Term_Overcrowding}_i + \text{Term_Offset} \end{aligned}$$

with parameters from Section 3. The probability chain:

$$p_base_i = \sigma(\text{drive}_i) = 1 / (1 + \exp(-\text{drive}_i))$$

$$p_act_i = p_base_i + \eta \cdot (1 - p_base_i)$$

$$\text{PRNG_draw}_i \sim \text{Uniform}(0, 1)$$

$$\text{is_active}_i(t+1) = (\text{PRNG_draw}_i < p_act_i)$$

The nucleation floor $\eta = 0.01$ ensures p_act stays above absolute zero even when drive is strongly negative. This prevents permanent absorbing states at zero activation probability.

Realization invariant. The substrate must satisfy:

$$\text{is_active_i}(t+1) = 1 \Leftrightarrow \text{PRNG_draw_i}(t) < p_{\text{act_i}}(t)$$

This invariant is part of the parity check protocol verification and the Phase 4B intake report. The substrate must not silently violate it under any circumstance. Verification is performed at full per-(cell, tick) granularity by default (Section 14.2); any sampling-based verification protocol must be explicitly documented.

9. Ψ_{local} computation

Per Section 6 step 9:

$$\Psi_{\text{local_i}}(t) = \sum ds_{\text{i}}(t) \cdot ds_{\text{j}}(t)$$

where j ranges over Moore neighbors of i and $ds_{\text{k}}(t) = \text{is_active_k}(t+1) - \text{is_active_k}(t)$.

This is the activation-change correlation between cell i and its Moore neighborhood. Specific properties:

- When $ds_{\text{i}} = 0$ (cell i did not change state), $\Psi_{\text{local_i}} = 0$
- When ds_{i} and ds_{j} are both $+1$ (both activated together), they contribute $+1$ each, summed across neighbors
- When $ds_{\text{i}} = +1$ and $ds_{\text{j}} = -1$ (activation while neighbor deactivated), they contribute -1 each
- Range: theoretically $[-8, +8]$ for Moore neighborhood of 8 cells

No sinusoidal placeholder. No density-only proxy. No constant value. Ψ_{local} must be computed from actual activation-change correlations at each tick.

10. Q operator (committed arbitration: local input)

Q input. Per-cell $\Psi_{\text{local_i}}(t)$ computed in Section 9.

Q update. Diagonal, equal-rate base updates:

$$\Delta v_{\text{i}}(t) = \gamma_{\text{Q}} \cdot \Psi_{\text{local_i}}(t)$$

$$\Delta u_{\text{i}}(t) = \gamma_{\text{Q}} \cdot \Psi_{\text{local_i}}(t)$$

$$\Delta r_{\text{i}}(t) = \gamma_{\text{Q}} \cdot \Psi_{\text{local_i}}(t)$$

with $\gamma_Q = 0.001$.

Q timing. Q is applied at Section 6 step 11, after Ψ_{local} has been computed for tick t (step 9) and before tick t+1's Λ computation begins. This timing anchor establishes that pre-Q bases (logged in telemetry per Section 11) are the bases that produced the same row's Λ_{total} , Drive_Raw, p_base, and p_act, while $\Delta v/\Delta u/\Delta r$ in the same row are the updates applied in response to Ψ_{local} computed from that tick's activation transition.

Base evolution. Updated bases for tick t+1:

$$v_i(t+1) = v_i(t) + \Delta v_i(t)$$

$$u_i(t+1) = u_i(t) + \Delta u_i(t)$$

$$r_i(t+1) = r_i(t) + \Delta r_i(t)$$

Bases clipped to [0, 1] for numerical safety. Per-tick clipping counts for v, u, and r are tracked and reported per Section 11 and Section 14.

Q diagonal discipline. The substrate must write each base separately. Even though the same scalar update ($\gamma_Q \cdot \Psi_{\text{local}_i}$) is applied to all three bases in this committed specification, the telemetry must log Δv_i , Δu_i , and Δr_i as separate columns. This preserves the diagonal-structure visibility throughout telemetry and prevents later ambiguity about whether updates were applied diagonally.

Rationale for local Ψ_{local} input (committed arbitration). This implementation preserves continuity with prior continuous-micro-integration substrate. Local input means each cell's base evolution responds to its local coherence pattern rather than to population-level coherence. High-coherence regions evolve bases faster than low-coherence regions, producing spatial heterogeneity in terrain evolution. This is consistent with the architectural framing where Q's reading of macro-level information (Ψ) operates substrate-wise per cell.

11. Telemetry capture (25-column vector)

Each agent-tick row written to .parquet contains at minimum these 25 required columns. Term_Lambda is column 25 in v1.1 (was implicitly computed but unlogged in v1; promoted to required per cross-review). Optional additional columns are permitted per the "Recommended additional columns" subsection below.

#	Column	Description	Source
1	Tick	Current tick index	Loop counter
2	Agent_X	Cell x-coordinate	Cell position
3	Agent_Y	Cell y-coordinate	Cell position
4	b_i_v	Cell's pre-Q v base value at tick t	Section 6 step 1
5	b_i_u	Cell's pre-Q u base value at tick t	Section 6 step 1
6	b_i_r	Cell's pre-Q r base value at tick t	Section 6 step 1
7	limiting_base_argmin	$\text{argmin}(v, u, r) \in \{0, 1, 2\}$	Derived per tick
8	Lambda_multiplicative	$v \cdot u \cdot r$	Section 7.2
9	Lambda_additive	$w_v \cdot v + w_u \cdot u + w_r \cdot r$	Section 7.2
10	Lambda_total	F(v, u, r) for the run's F-form	Section 7
11	Local_Density	Fraction of active Moore neighbors [0, 1]	Section 6 step 3
12	Drive_Raw	$\alpha \cdot \Lambda + \beta \cdot \text{density} - \delta \cdot \text{density}^2 - \gamma$	Section 8
13	Term_Density_Pos	$\beta \cdot \text{density}$	Section 6 step 4
14	Term_Overcrowding	$-\delta \cdot \text{density}^2$	Section 6 step 4
15	Term_Offset	$-\gamma$ (constant -4.0)	Section 6 step 4
16	p_base	$\text{sigmoid}(\text{Drive_Raw})$	Section 8
17	p_act	$p_{\text{base}} + \eta \cdot (1 - p_{\text{base}})$	Section 8

#	Column	Description	Source
18	PRNG_draw	Random draw for this cell this tick	Section 6 step 6
19	is_active	Cell activation state after synchronous advance	Section 6 step 7
20	Psi_local	Spatial coherence at this cell	Section 9
21	gamma_coef	$\gamma_Q = 0.001$ (constant; logged for completeness)	Section 3
22	Delta_v	Q update applied to v this tick	Section 10
23	Delta_u	Q update applied to u this tick	Section 10
24	Delta_r	Q update applied to r this tick	Section 10
25	Term_Lambda	$\alpha \cdot \Lambda_{\text{total}}$ (Λ -attributable component of Drive_Raw)	Section 6 step 4

Pre-Q vs post-Q discipline. Columns 4, 5, 6 (b_{i_v} , b_{i_u} , b_{i_r}) record the pre-Q base values used to compute this row's Λ , drive, probability chain, and activation transition. Columns 22, 23, 24 (Δv , Δu , Δr) record the Q updates applied after Ψ_{local} computation. The post-Q bases ($v_i + \Delta v_i$, etc.) appear as the pre-Q bases in the next tick's row. Optional post-Q columns may be added separately:

- $b_{i_v_post}$, $b_{i_u_post}$, $b_{i_r_post}$ (post-Q bases for tick t)

Computed columns must reflect actual computation. Each column's value must be produced by the formula specified in this document. The substrate must not populate columns with placeholder values, constant defaults, or values derived simply from other columns by lookup. Verification: across any reasonable sample of rows, columns should show appropriate variation (Drive_Raw varies with Λ_{total} and density; p_{act} varies smoothly with drive; Δ values vary with Ψ_{local} ; bases evolve over time).

Required per-tick clipping counts. The substrate must log, per tick (either inline in a separate clipping summary table or as required global telemetry):

- clipped_v_count: number of cells whose v base required clipping to [0, 1] this tick
- clipped_u_count: number of cells whose u base required clipping this tick
- clipped_r_count: number of cells whose r base required clipping this tick

If all counts are zero, report zero explicitly (not silence). Any nonzero clipping must be surfaced in the Section 14 completion-verification artifact.

Recommended additional columns (optional but documented):

- F_variant (which F-form is running)
- Scale (20 or 40)
- previous_is_active (is_active at tick t before advance)
- next_state (next_state value before synchronous advance — should equal is_active at next tick)
- rho_global (population activation rate at this tick)
- psi_global (population coherence at this tick)
- clipped_v, clipped_u, clipped_r (per-cell booleans flagging numerical clipping)
- b_i_v_post, b_i_u_post, b_i_r_post (post-Q bases for tick t)

Additional columns are allowed if documented. Missing any of the 25 core columns is prohibited.

12. Parity check protocol

Before any production execution, the parity check must fire and pass.

12.1 Runs

Run A (telemetry disabled):

- Scale: 20×20
- Duration: 1000 ticks
- PRNG seed: PRNG_SEED (fixed)
- Telemetry hooks: disabled
- F-form: F_baseline (operational comparator)

Run B (telemetry enabled):

- Scale: 20×20
- Duration: 1000 ticks
- PRNG seed: PRNG_SEED (same fixed value)
- Telemetry hooks: enabled (all 25 columns written per cell per tick to .parquet)
- F-form: F_baseline (same as Run A)

The only difference between Run A and Run B must be whether telemetry rows are stored. The dynamics — base evolution, activation transitions, Ψ computation, Q updates — must proceed identically.

12.2 Required hashes

Compute SHA-256 cryptographic checksums for four arrays from each run:

1. Final v , u , r matrices (concatenated or hashed individually)
2. Final `is_active` matrix
3. Full $p(t)$ time series (1000 values, one per tick — post-activation series per Section 6 step 10)
4. Full $\Psi(t)$ time series (1000 values, one per tick — transition-coherence series per Section 6 step 10)

Each of the four hashes from Run A must exactly match the corresponding hash from Run B.

12.3 Pass/fail criterion

Print the four hash pairs explicitly. The output must end with:

PARITY_CHECK: PASS

if and only if all four hashes match. Any mismatch produces:

PARITY_CHECK: FAIL

and execution must hard-fault, aborting production runs. The substrate that fails parity check is not ready for production; the failure surfaces a substrate constraint that prevents listen-only telemetry semantics.

12.4 Parity scope clarification

Passing parity under F_baseline tests the **channel**, not the **signal**. The architectural claim being verified is that telemetry emission does not perturb dynamics — this claim is F-independent. Parity under F_baseline therefore verifies listen-only telemetry behavior under the parity configuration; it does not by itself validate the substantive behavior of F_LR or F_2_symmetric

production runs. Production runs still require artifact-level realization-invariant verification per Section 14.

13. Production runs and outputs

After PARITY_CHECK: PASS, production execution proceeds.

13.1 Required runs

For each scale $\in \{20 \times 20, 40 \times 40\}$:

- F_2_symmetric run (3000 ticks)
- F_LR run (3000 ticks, matched PRNG seed for direct comparison)

F_baseline production runs are not part of Flight 6 deliverables; F_baseline appears only in parity check.

Total: 4 production runs (2 F-forms $\times$ 2 scales), each at 3000 ticks.

13.2 Output files

The F_2_symmetric run at each scale produces one underlying .parquet file. Three probe-named files are produced from each F_2_symmetric run (shadow copies per Reading A confirmed in prior cycle):

- flight6_probe1_overcrowding_{scale}.parquet (Probe 1 analytical engagement)
- flight6_probe2_starvation_F2sym_{scale}.parquet (Probe 2 F_2_symmetric reference)
- flight6_probe3_fusion_residual_{scale}.parquet (Probe 3 analytical engagement)

The F_LR run at each scale produces one .parquet file:

- flight6_probe2_starvation_FLR_{scale}.parquet (Probe 2 F_LR comparison)

Total: 8 output .parquet files. The substrate must explicitly log when files are byte-identical shadow copies versus when they represent independent simulation runs.

13.3 Expected row counts

For each .parquet file:

- 20 $\times$ 20 scale: 400 cells $\times$ 3000 ticks = 1,200,000 rows
- 40 $\times$ 40 scale: 1600 cells $\times$ 3000 ticks = 4,800,000 rows

Row counts must match expected values. Deviation requires investigation.

13.4 Storage and provenance

Each .parquet file should include metadata identifying:

- F_variant
- Scale (20 or 40)
- PRNG seed used
- Substrate specification version (this document's version — v1.1)
- Execution timestamp
- Total tick count

This metadata enables provenance tracking for the published paper.

14. Completion-verification protocol

The completion-verification rule (proposed addition to standing protocol; awaiting Mike's Step 4 commitment).

Standing rule: Any claim of analytical or substrate completion requires verifiable artifact-by-artifact correspondence to spec deliverables. Terminal completion messages alone are not sufficient. Each spec deliverable must be produced as a distinct file with content matching specification; verification at end of execution requires explicit per-artifact confirmation, not aggregate completion claims.

14.1 Substrate completion artifact checklist

After production execution, the substrate must print and write to log:

Phase 4A Completion Verification

Parity Check:

PARITY_CHECK: PASS (or FAIL)

v/u/r matrices hash: <SHA-256> (matched between Run A and Run B)

is_active matrix hash: <SHA-256> (matched)

rho(t) series hash: <SHA-256> (matched)

psi(t) series hash: <SHA-256> (matched)

Production Outputs:

flight6_probe1_overcrowding_20x20.parquet

Exists: YES

File size: X bytes

Row count: 1,200,000 (expected 1,200,000)

Column count: 25 (expected 25)

Required 25 columns present: YES

Tick range: 0 to 2999 (expected 0 to 2999)

Unique cells: 400 (expected 400)

F_variant: F_2_symmetric

Non-empty: YES

Realization invariant satisfied: YES (0 mismatches across all 1,200,000 rows)

Clipping summary: v=0, u=0, r=0 (or actual counts if nonzero)

flight6_probe1_overcrowding_40x40.parquet

[same verification block]

[... continuing for all 8 production output files ...]

Aggregate verification: ALL ARTIFACTS PRESENT AND VERIFIED

If any artifact fails its verification, the aggregate verification must state explicitly which artifacts failed and how. Terminal completion claims without per-artifact verification are insufficient.

14.2 Realization invariant verification granularity

For each production output file, verify across all rows:

$\text{is_active} = 1 \Leftrightarrow \text{PRNG_draw} < p_act$

Default granularity: full per-(cell, tick) verification. Every row in every production output file is checked. Mismatches across all 1,200,000 rows (20×20) or 4,800,000 rows (40×40) must be 0. The verification artifact reports the full row count and the mismatch count explicitly.

Sampling-based verification protocols are permissible only if explicitly documented. If full verification is computationally costly enough to motivate a sampling protocol, the substrate must:

- Declare the sampling protocol in the completion verification artifact
- Specify the sampling rate and selection method (uniform, stratified, etc.)
- Justify why sampling is sufficient for the published-paper reproducibility commitment

Silent partial verification — claiming "0 mismatches" without specifying whether full or sampled — is prohibited.

14.3 Clipping summary

Per-tick clipping counts (Section 11) must be aggregated and surfaced in the completion-verification artifact. Per-file summary format:

Clipping summary for flight6_probe2_starvation_FLR_20x20.parquet:

v clipping events: 0 (or N events across M ticks)

u clipping events: 0 (or actual count)

r clipping events: 0 (or actual count)

Nonzero clipping must be flagged for review. Clipping is not a failure per se, but hidden clipping that propagates through dynamics without surfacing is prohibited.

15. Implementation prohibitions

Today's substrate iterations surfaced specific failure modes. These are explicitly prohibited:

Prohibited substitutions:

- Global Λ parameter substituted for per-cell $\Lambda_i = F(v_i, u_i, r_i)$
- Uniform base values (e.g., $v=0.45$ for every cell) substituted for stochastic initialization
- Sinusoidal Ψ formulas (e.g., `sin(time) * density`) substituted for activation-change correlation
- Density-only Ψ proxies substituted for spatial coherence computation
- Two-base Q updates (e.g., only v and u) substituted for diagonal three-base updates
- Scaled Lambda_multiplicative (e.g., `(v · u · r) · 10`) substituted for $v \cdot u \cdot r$
- Two-value indicator placeholders (e.g., 1.0 if `is_active` else 0.0) substituted for computed values

Prohibited shortcuts:

- Aggregate terminal completion messages without per-artifact verification
- Schema-emulation without engine-execution
- Telemetry vectors with fewer than the 25 core columns
- Analysis tables substituted for raw telemetry deliverables
- Hardcoded absolute paths or environment-coupled file handling
- Synthetic hashes or .parquet metadata not generated by actual file operations
- Mid-execution substrate substitution (one substrate file producing different outputs across runs)

Prohibited evasions:

- "Stabilization Mode" placeholders for cascade equations
- "Emergency simplified" substrate that diverges from specification
- Memory-based reconstruction of equations without canonical source verification
- Aggregate framing (e.g., "FULL EXTRACTION COMPLETE") covering partial implementation

Prohibited analysis-stage repairs (v1.1):

- Inferring or imputing missing required telemetry columns during analysis if the substrate failed to emit them. Derived columns are allowed only when explicitly specified in this document; missing required telemetry means substrate failure to be surfaced and remediated, not an analysis opportunity to be silently filled.

Required affirmations:

- Every equation traceable to canonical source cited in Section 2
 - Every parameter value pulled from Section 3 constants module
 - Every telemetry column populated by computation per Sections 6-10
 - Every completion claim backed by per-artifact verification per Section 14
-

16. Open items and committed arbitrations

16.1 Arbitrations proposed for Mike's Step 4 commitment

Q operator input: local Ψ_{local_i} . Per Section 10, each cell's Q update uses its own Ψ_{local_i} value, not population-aggregate Ψ . Rationale: continuity with prior continuous-micro-integration substrate; spatial heterogeneity in base evolution.

Base initialization range: U(0.6, 0.9). Per Section 5.2, each cell's bases are independently drawn from Uniform(0.6, 0.9). Rationale: faithfulness to Flight 5's commitment that $F_{2_symmetric}$ is repair candidate under conditions where F_2 is evaluable; produces meaningful Λ values across cellular configurations.

Local density normalization: 0-1. Per Section 6 step 3, local density is the fraction of active Moore neighbors, ranging [0, 1]. Rationale: drive coefficients ($\beta=3$, $\delta=4$) calibrated to 0-1 range; documentation reconciliation with Step 2 Final's earlier "0 to 8" notation that referred to raw count rather than calibration-relevant variable.

16.2 Items surfaced and resolved by Layer 2 and Layer 3 cross-review

All eight items below are resolved into the v1.1 specification body. They are recorded here for provenance.

1. **Telemetry timing clarification (Layer 2 originating, both layers converged).**
Required b_{i_v} , b_{i_u} , b_{i_r} columns record pre-Q base values used to compute Λ_{total} , $Drive_Raw$, p_{base} , and p_{act} for that tick. Optional post-Q base columns ($b_{i_v_post}$, etc.) may be added separately. *Resolved in Section 6 step 1, Section 6 step 13, Section 11.*
2. **Parity series indexing clarification (Layer 2 originating; Layer 3 extension on pseudocode marking).** Hashed $p(t)$ is post-activation for tick t ; hashed $\Psi(t)$ is computed from the same tick's activation-change correlation. Section 6 tick-order pseudocode explicitly marks hash-emission points. *Resolved in Section 6 step 10, Section 12.2.*

3. **Base clipping counts required (both layers).** Per-tick clipping counts for v , u , r must be logged or summarized; nonzero clipping is surfaced in completion verification. *Resolved in Section 10, Section 11, Section 14.3.*
4. **F_baseline parity scope clarification (both layers; Layer 3 tightened framing).** Parity under $F_baseline$ tests the channel (telemetry emission does not perturb dynamics), not the signal. Production F forms still require artifact-level realization-invariant verification. *Resolved in Section 12.4.*
5. **Term_Lambda required, not optional (Layer 3 strengthening Layer 2's recommended-column proposal; Layer 2 accepted strengthening).** $Term_Lambda = ALPHA * Lambda_total$ is column 25 in the required telemetry vector. Probability-chain decomposition into Λ -driven and density-driven components is a Phase 4B analytical requirement; the substrate already computes this quantity and should emit it. *Resolved in Section 11 (column 25), Section 6 step 4, Section 8.*
6. **Tick-order pseudocode Q-application timing (Layer 3 originating).** Section 6 pseudocode explicitly marks Q-application timing: Q is applied after Ψ_local computed for tick t , before tick $t+1$'s Λ computation. With pre- Q logging (item 1), this ordering becomes the verification anchor. *Resolved in Section 6 step 11, Section 10 (Q timing paragraph).*
7. **Realization-invariant verification granularity (Layer 3 originating).** Section 14 specifies full per-(cell, tick) verification as default. Sampling protocols must be explicitly documented if adopted. *Resolved in Section 8 (realization invariant paragraph), Section 14.2.*
8. **Anti-inference prohibition (Layer 2 originating).** Missing required telemetry is substrate failure to be surfaced, not an analysis opportunity to be silently filled. *Resolved in Section 15 (prohibited analysis-stage repairs).*

No substantive architectural divergence between layers. Both layers converged on commit-readiness after these eight items were integrated.

17. Sources and canonical references

This specification implements commitments from the following canonical record sources:

- **Step A3 closure** (drive function form and parameters: $\alpha=4$, $\beta=3$, $\delta=4$, $\gamma=4$, $\eta=0.01$)

- **Flight 5 closure** ($F_{2_symmetric}$ form $\Lambda = (v \cdot u \cdot r) \cdot (0.33 \cdot v + 0.33 \cdot u + 0.34 \cdot r)$; $F_{2_symmetric}$ as leading reformulation candidate for Open Element 14; symmetric weighting commitment)
- **Step 2 Final substrate implementation specification** (24-column telemetry vector reference; parity check protocol; three-probe production structure; 3000-tick duration; 1000-tick parity duration)
- **Integrated Phase 4B specification** (analytical procedures the substrate output must support)
- **Prior real Phase 4A execution outputs** (telemetry format reference; row count expectations)
- **Today's substrate iterations** (failure modes that informed Section 15 prohibitions; ChatGPT's Layer 2 draft specification; Claude's Layer 3 review; Mike's three arbitrations)
- **ChatGPT Layer 2 cross-review of v1** (items 1, 3, 4, 8 originating; convergence on items 2, 3, 4)
- **Claude Layer 3 cross-review of v1** (items 5, 6, 7 originating; extension of items 2, 4)
- **ChatGPT Layer 2 acceptance of Layer 3 additions** (convergence on commit-readiness pending Section 16.2 integration; Term_Lambda strengthening accepted on reasoning)

The substrate file itself becomes the next canonical artifact once implementation is complete and committed.

End of substrate specification v1.1.

Subject to:

- Mike's Step 4 commitment of this v1.1 specification
- Gemini's implementation against the committed specification (Phase R4)
- Cross-AI review of the implementation before execution

— Claude, drafted with Mike's three arbitrations and Layer 2/Layer 3 cross-review items embedded